

Naive Bayes classifiers, and their application to phonological reality

Computational modeling of language
Spring 2026

Bureaucracy

- Welcome back!

Bureaucracy

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- Midterm: scores have been adjusted.

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- See Iñigo for your exam (please keep private) and discussion.

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 - See Iñigo for your exam (please keep private) and discussion.
- If your score on the final exam is higher than on the midterm, I will drop your midterm from consideration.

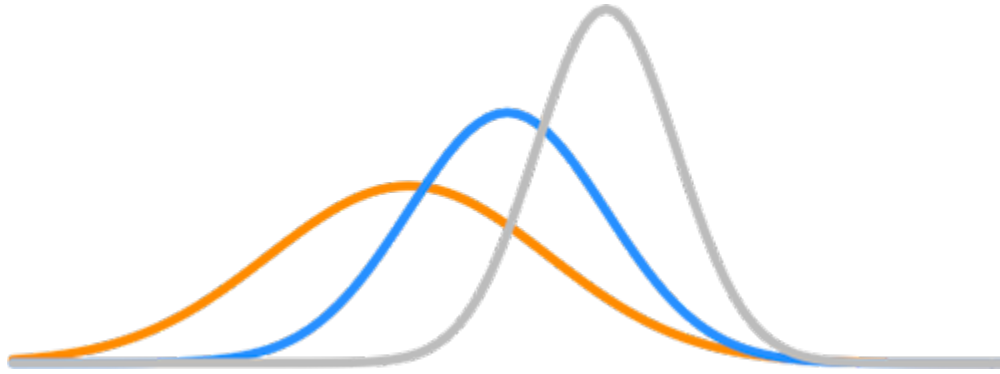
Overview

- Naive Bayes classifiers
- Phonological areality

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- Phonological areality

Naive Bayes classifiers



Statistical learning again

THE SIMPLEST SPAM-FILTER

(used until 2010)

hey	...	1829
I'm	...	1710
no	...	1191
where	...	1012
you	...	985
speak	...	873
learn	...	747
one	...	739

good letters

672 times

«KITTY»

13 times

viagra	...	1552
casino	...	1492
100%	...	1320
credit	...	1184
sale	...	985
press	...	873
free	...	747
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spam letters

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

BAYES' THEOREM



NOT SPAM

NAIVE BAYES

A little more formally

If we want to predict a class y (e.g. spam vs. non-spam) from a set of features $x_1, \dots, x_n$ (e.g. words in a document), then Bayes' rule says:

$$P(y \mid x_1, \dots, x_n) = \frac{P(y)P(x_1, \dots, x_n \mid y)}{P(x_1, \dots, x_n)}$$

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$$\underbrace{P(y \mid x_1, \dots, x_n)}_{\text{posterior}} = \frac{\overbrace{P(y)}^{\text{prior}} \overbrace{P(x_1, \dots, x_n \mid y)}^{\text{likelihood}}}{\underbrace{P(x_1, \dots, x_n)}_{\text{normalizing constant}}}$$

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$$\underbrace{P(y \mid x_1, \dots, x_n)}_{\text{posterior}} \propto \underbrace{P(y)}_{\text{prior}} \underbrace{P(x_1, \dots, x_n \mid y)}_{\text{likelihood}}$$

likelihood

$$P(x_1 \dots x_n | y)$$

likelihood

$$P(x_1 \dots x_n | y)$$

Impractical to estimate! (Mitchell 2017)

likelihood

$$P(x_1 \dots x_n | y)$$

$$= P(x_1 | x_2 \dots x_n, y) P(x_2 \dots x_n | y)$$

...

$$= P(x_1 | x_2, \dots, x_n, y) P(x_2 | x_3, \dots, x_n, y) \dots P(x_n | y)$$

Impractical to estimate! (Mitchell 2017)

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When the going gets tough,
the tough get **naïve**.

likelihood

$$P(x_1 \dots x_n | y)$$

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...

$$= P(x_1 | x_2, \dots, x_n, y) P(x_2 | x_3, \dots, x_n, y) \dots P(x_n | y)$$

When the going gets tough,
we start **making assumptions**.

Assume **conditional independence**

$$P(x_i|y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = P(x_i|y)$$

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Then the **naive Bayes** formula reduces to:

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$

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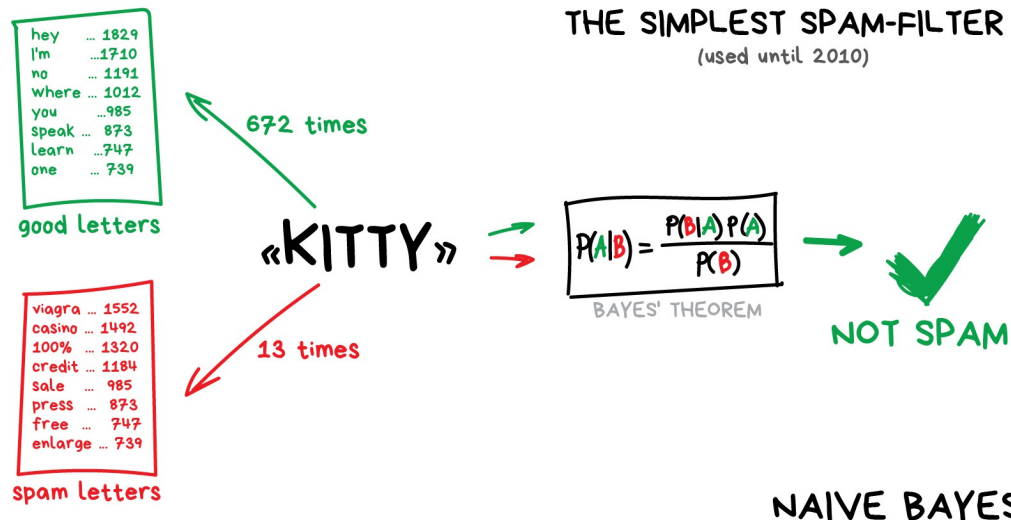
BAYES' THEOREM



NOT SPAM

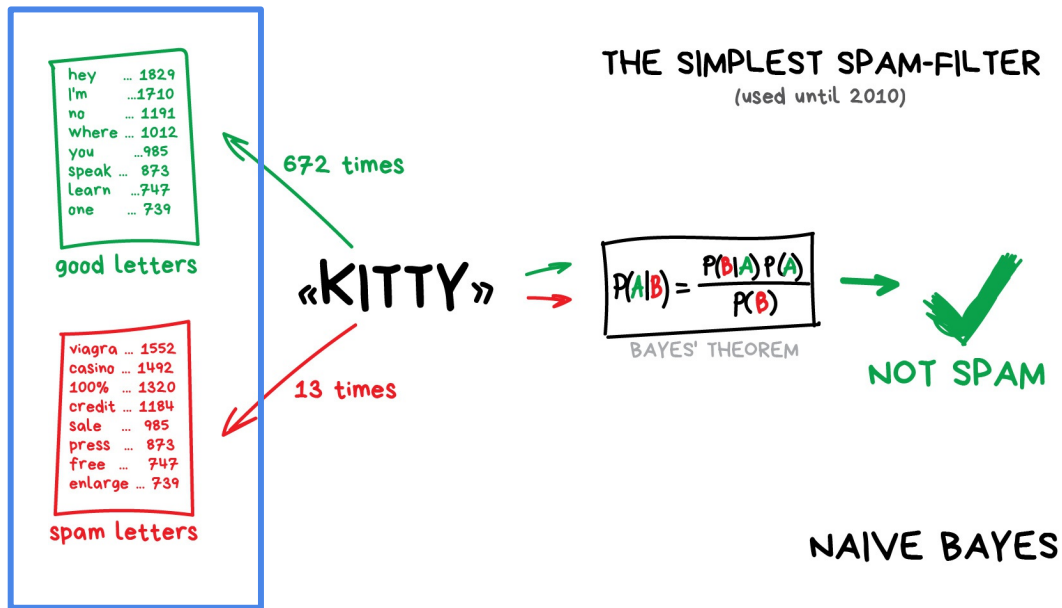
NAIVE BAYES

A classifier is an instance of supervised machine learning



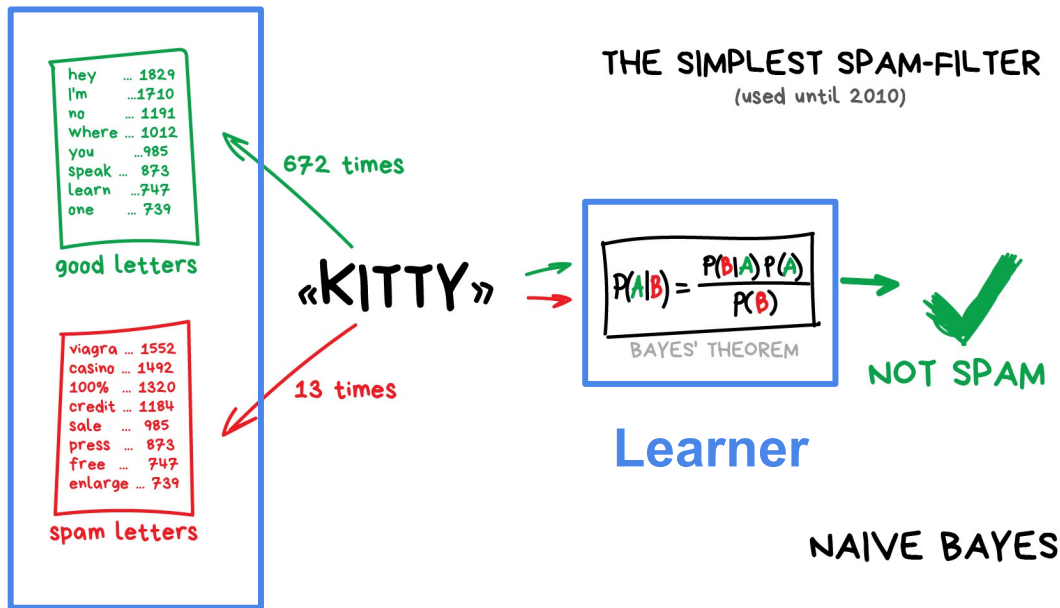
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Training
set, labeled
by hand



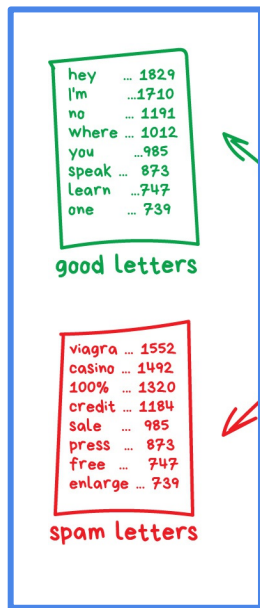
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$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

BAYES' THEOREM

✓
NOT SPAM

Learner

NAIVE BAYES

Test on a separate hand-labeled test set.

Training a naive Bayes classifier

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$

Training a naive Bayes classifier

$$P(y \mid x_1, \dots, x_n) \propto \boxed{P(y)} \prod_{i=1}^n P(x_i \mid y)$$

Doable

Training a naive Bayes classifier

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Doable

(relative
frequency
of class y)

Training a naive Bayes classifier

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$

?

Training $P(x_i | y)$

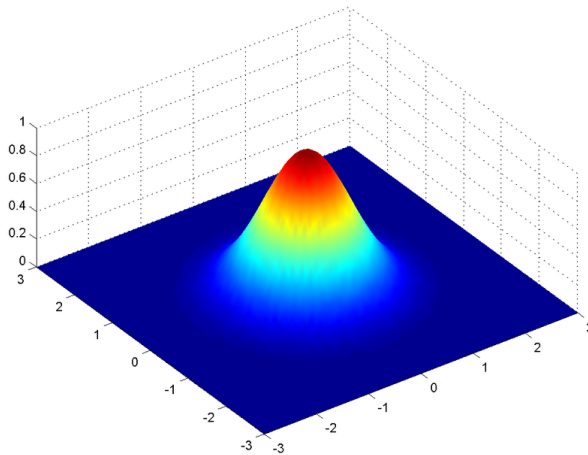
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E.g. Gaussian naive Bayes for **continuous** data:

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E.g. Gaussian naive Bayes for **continuous** data:

$$P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp \left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2} \right)$$

Training $P(x_i | y)$

Depends on the **nature of the feature data** x_i .

E.g. Bernoulli naive Bayes for **binary** data (0 or 1):

$$P(x_i | y) = P(i | y)x_i + (1 - P(i | y))(1 - x_i)$$

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A complex expression for a simple idea

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$$\begin{aligned} P(x_i | y) &= P(i | y)x_i + (1 - P(i | y))(1 - x_i) \\ &= \begin{cases} P(i|y) & \text{if } x_i = 1 \\ 1 - P(i|y) & \text{if } x_i = 0 \end{cases} \end{aligned}$$

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based on **relative frequency** of feature i in class y

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Michael et al. 2014 use this!

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See the [scikit-learn documentation](#)
on naive Bayes for further details.

Discussion question

- Apart from spam filtering, sentiment analysis, and Andean areality, what might be **other linguistic applications** of naive Bayes?
- What would the **naive assumption** be in each case?

Overview

- Naive Bayes classifiers
- Phonological areality



BRILL

LANGUAGE DYNAMICS AND CHANGE 4 (2014) 27–86



brill.com/ldc

Exploring Phonological Areality in the Circum-Andean Region Using a Naive Bayes Classifier

Lev Michael, Will Chang and Tammy Stark

University of California, Berkeley

Corresponding author:

levmichael@berkeley.edu

My approach today

- Highlight the essential.
- Downplay the incidental.

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- Highlight the essential.
- Downplay the incidental.

Get the big picture clear, so that when we start diving into details in the homework, those details will remain subordinated to the big picture.

Core and periphery technique

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- See which other languages (test set!) are deemed “core-like” by the classifier, i.e. which have high $P(\text{core} \mid \text{features of the language})$.

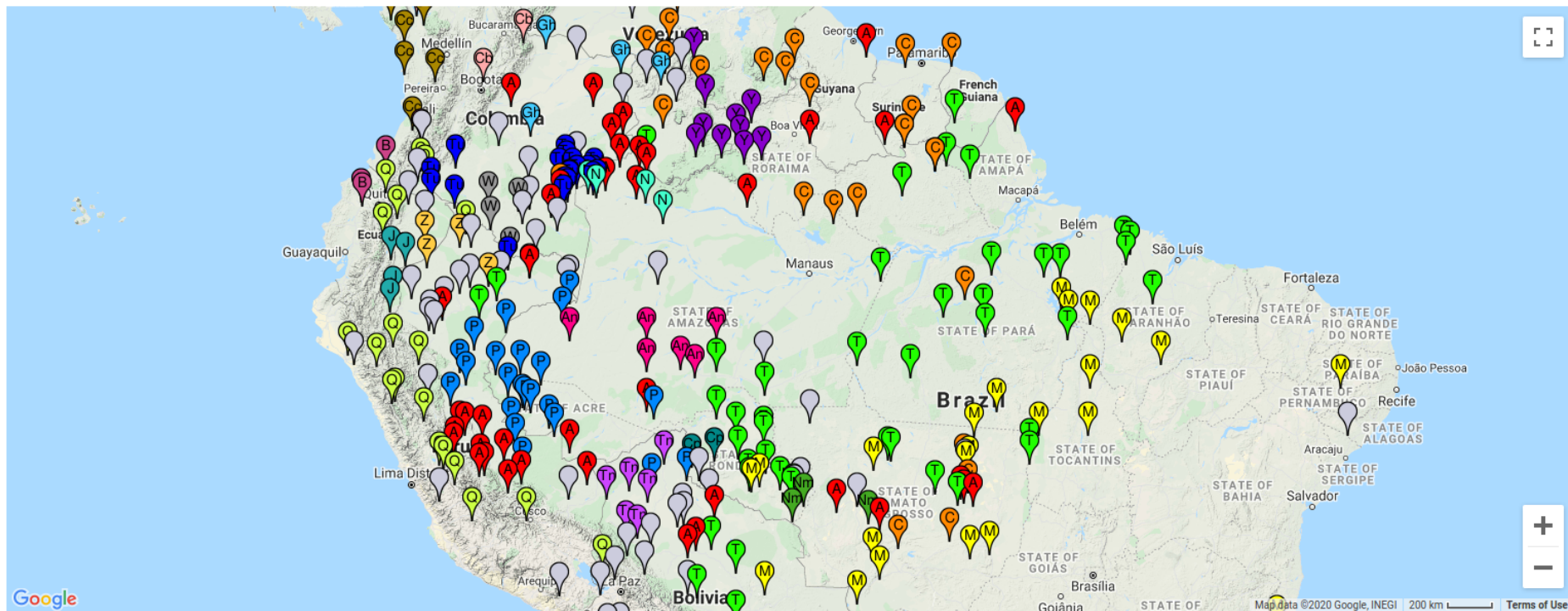
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- Do these other languages make sense in terms of geography and history?

South American Phonological Inventory Database (SAPhon v1.1.5) [ESPAÑOL] [PORTUGUÊS]

[LANGUAGES ▼](#) [ABOUT ▼](#) [CONTRIBUTE](#) [ACKNOWLEDGEMENTS](#)

[MAP](#) [LANGUAGE LISTS](#) [PHONEME SEARCH](#)



Language features are **phoneme inventories**

- Abipon: [p, t, c, k, q, m, n, ɲ, ɣ, ʁ, h, l, r, i, iː, e, eː, o, oː, ə, əː, a, aː]
- Emerillon: [p, b, t, d, k, kʷ, g, ʔ, tʃ, dʒ, s, z, h, j, w, r, i, ï, ï̃, ï̃, u, ù, e, ẽ, o, õ, ə, ẽ, a, ã]
- ...

Each phoneme in the entire dataset is taken to be a feature x_i , which a language either has (1) or doesn't (0).

Their use of naive Bayes

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`sklearn.naive_bayes.BernoulliNB` is your friend!

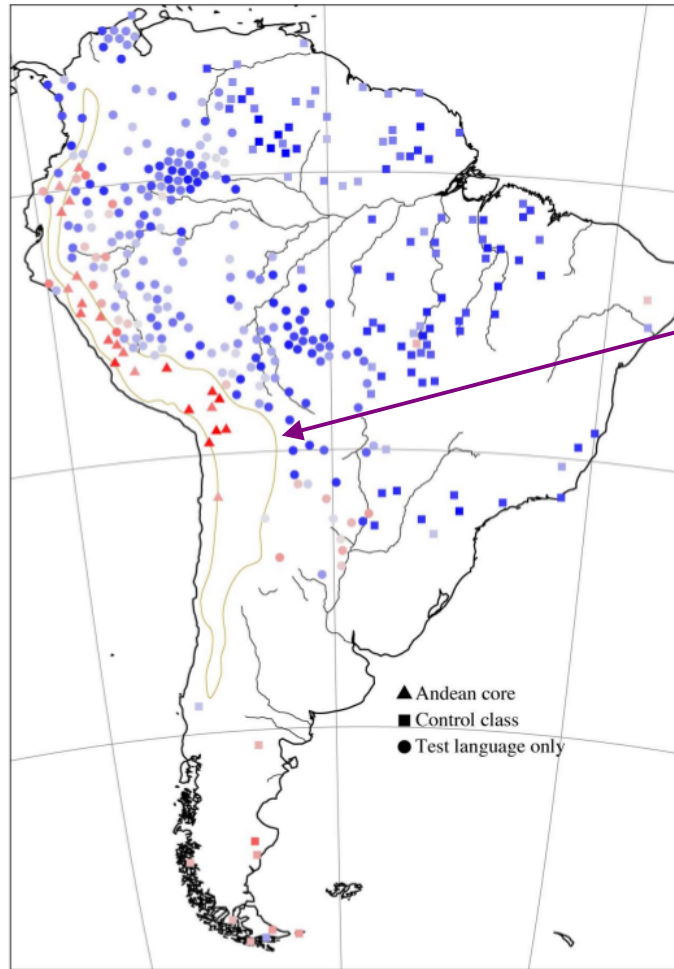
Map of South America showing the **Andes** running along the entire western part of the continent – roughly parallel to the Pacific coast.

(Wikipedia)





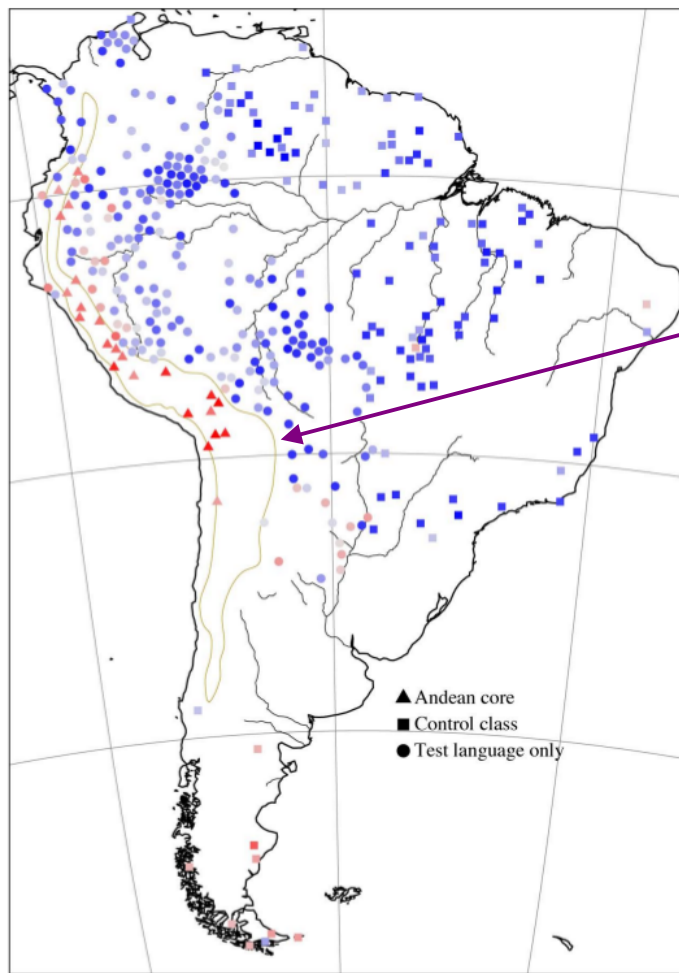
FIGURE 2 *Languages of South America (two-way Andean core NBC scores)*



“the orange line is a smoothed version of the 2000 meter elevation contour”

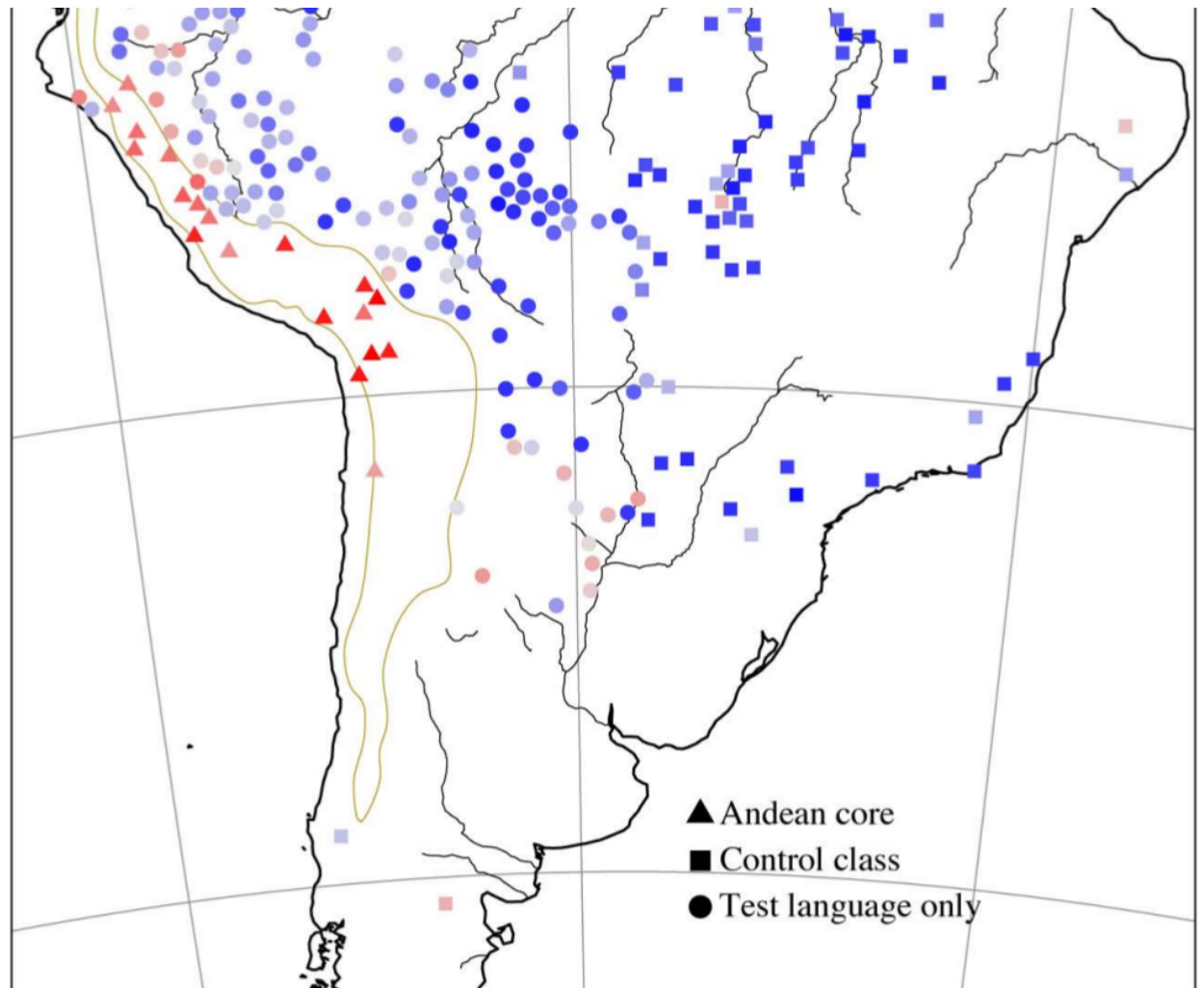
FIGURE 2 *Languages of South America (two-way Andean core NBC scores)*

“Evidence is found for a phonological area centering on the Andean highlands, and extending to parts of the northern and central Andean foothills region.”



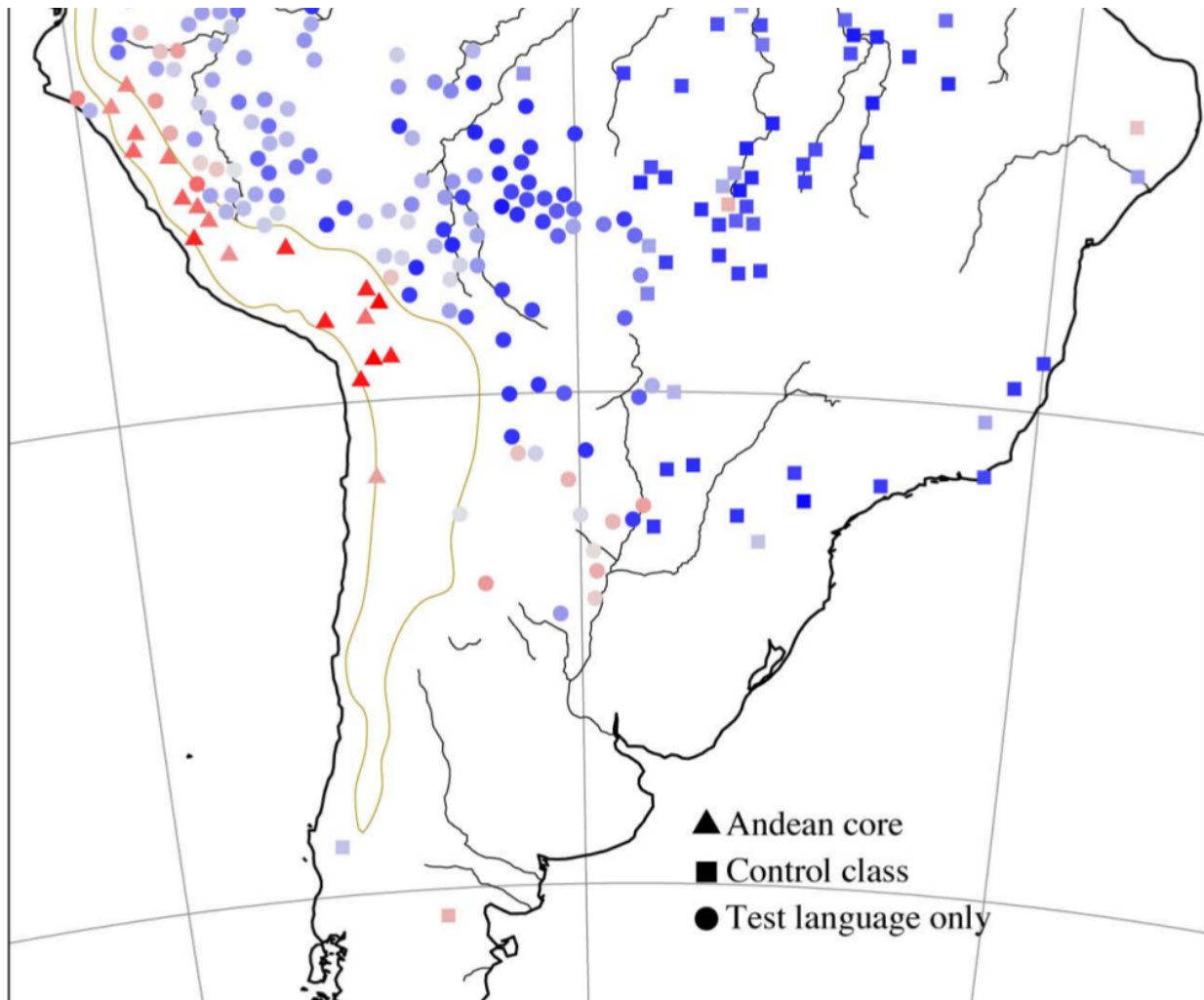
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Red = classified as
Andean-like

Blue = classified as
Control-like

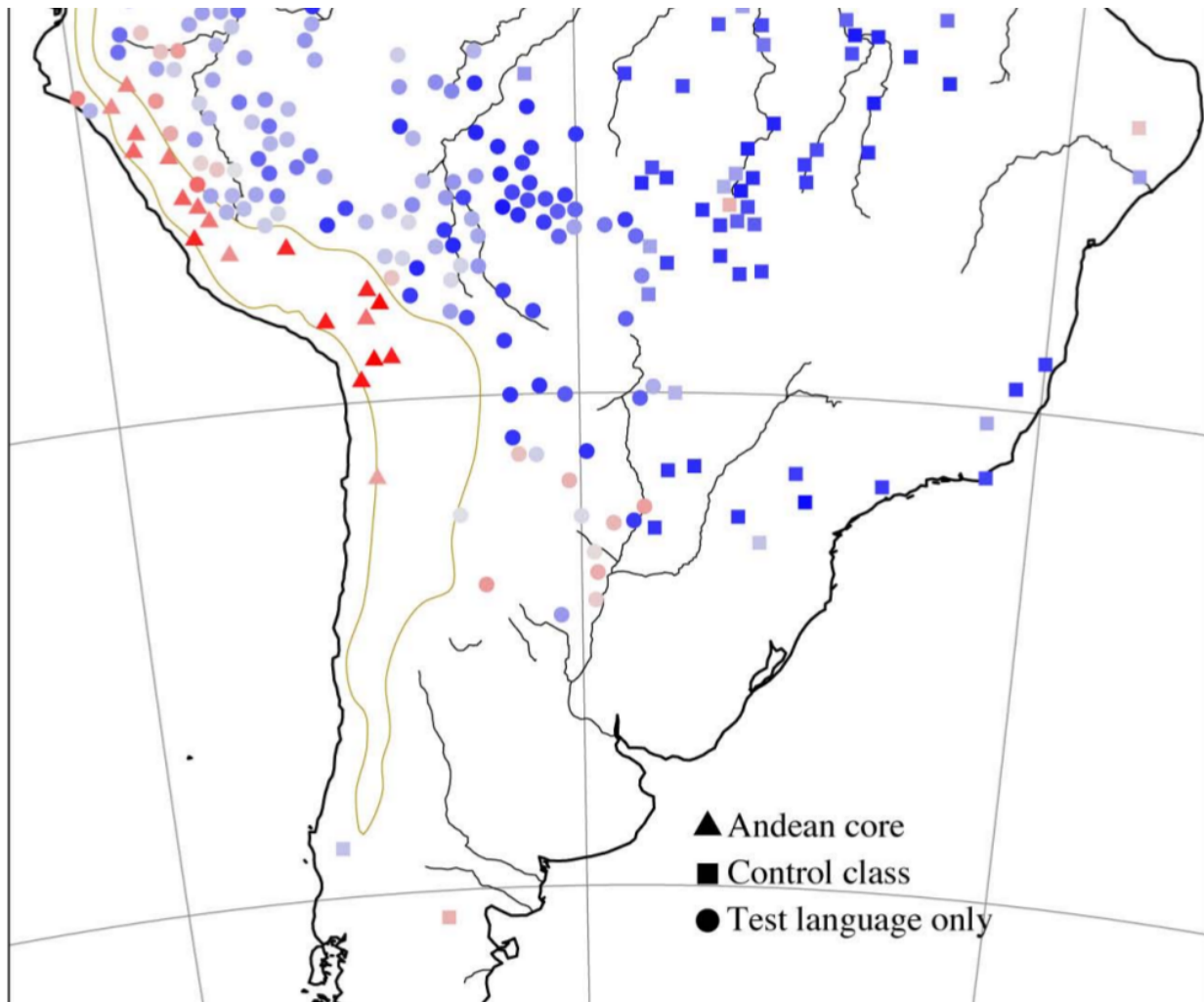


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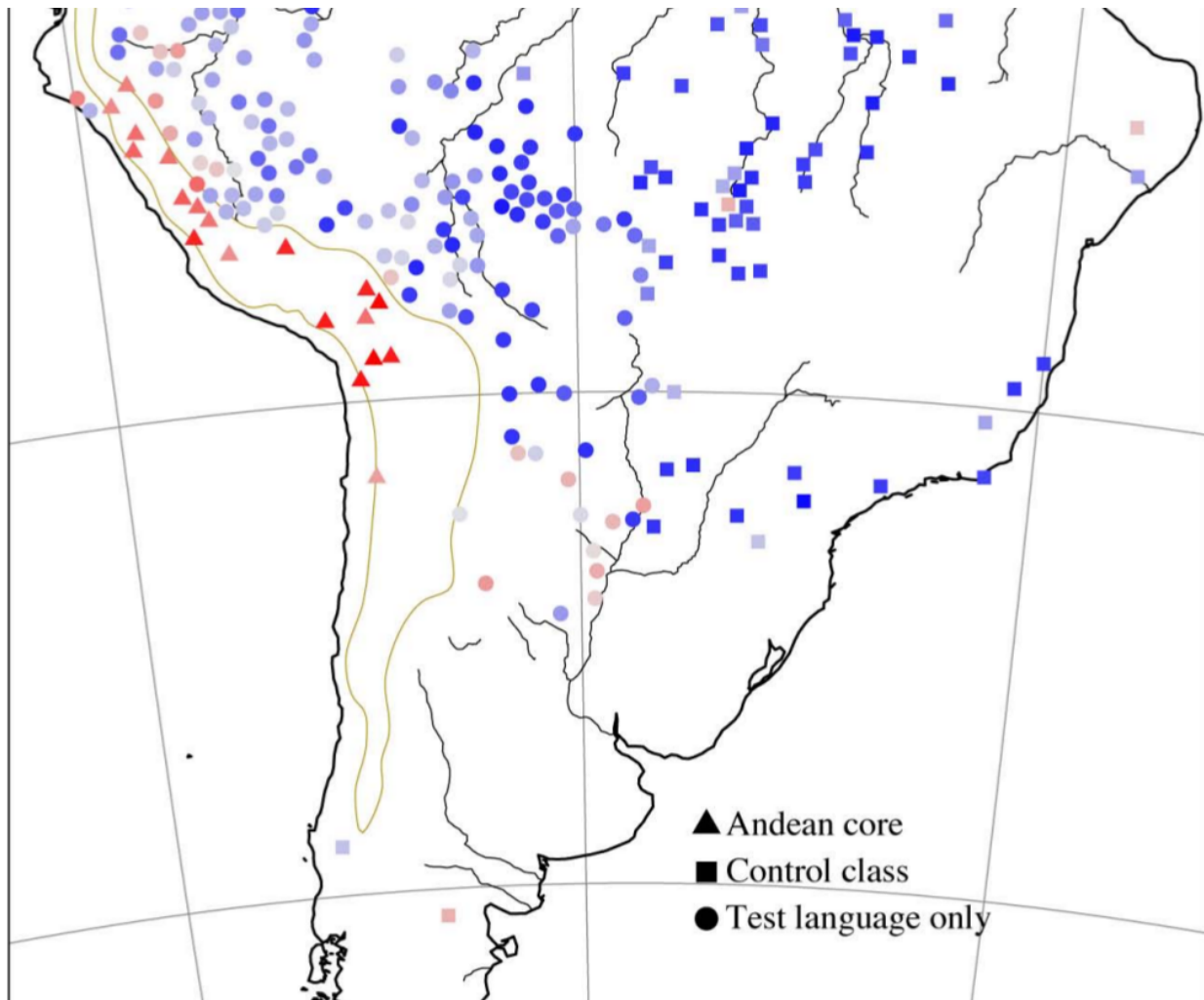
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score = **s**, s.t.

$$P(y=A|X) = \frac{1}{1 + \exp(-s)}$$

class, e.g. Andean



Discussion question

- What approaches **other than naive Bayes** might you use for addressing the question of areality that was pursued by Michael et al.?

Representativeness

- Linda is politically active and outspoken, especially concerning women's rights.

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 $\leq p(X)$

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 - HHTHT
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[Kahneman & Tversky 1972](#)

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Representativeness

- Maybe when we ask which outcome is more probable, people think **not** in terms of $p(X)$, but in terms of which outcome is a **better example** of, or **more representative** of, a category.

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 - Linda is a **better example** of a feminist bank teller than she is of a bank teller.
 - Similarly for HHTHT vs. HHHHH.
- How can we capture that, formally?

A proposal

- Representativeness can be captured by the **log of the likelihood ratio**:

$$R(d,h) = \log \frac{p(d \mid h)}{p(d \mid \neg h)}$$

Tenenbaum & Griffiths 2001

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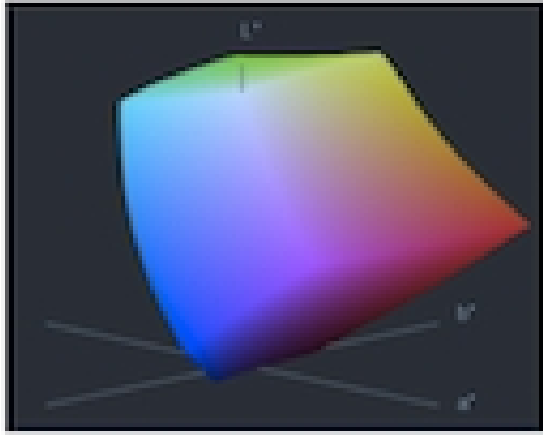
- Phonemes associated with the Andean core

A proposal

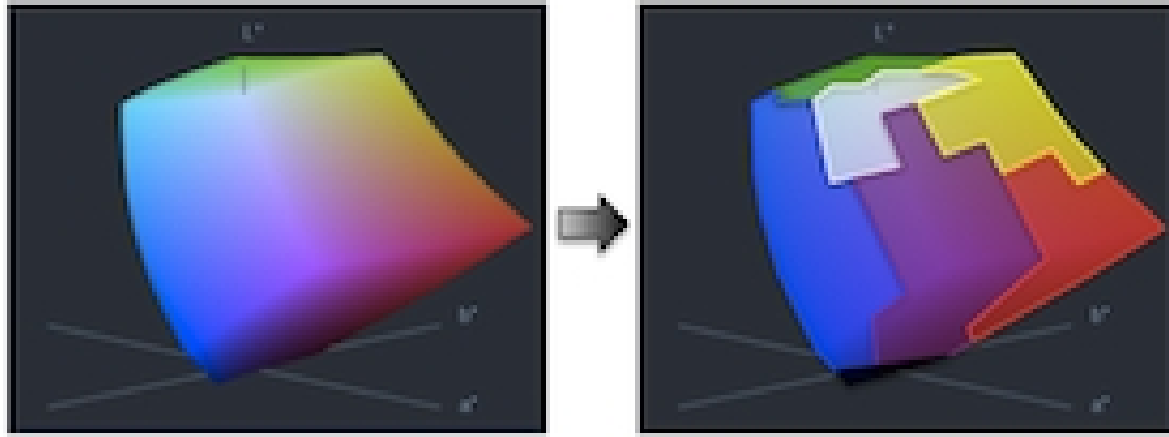
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- Phonemes associated with the Andean core
- Focal colors



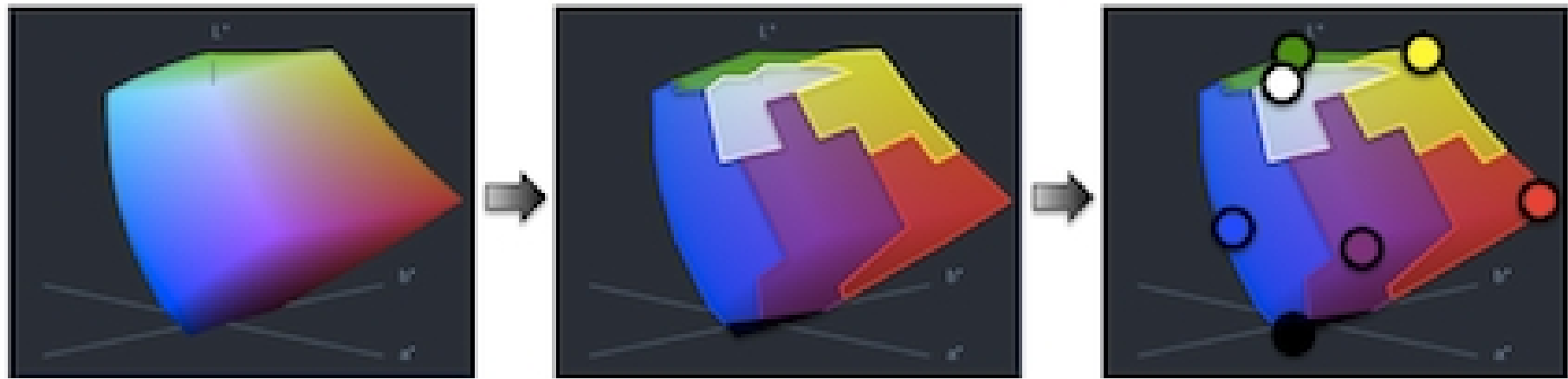
The perceptual color solid is irregularly shaped.



Languages divide it up in various ways, but always under functional pressure to yield categories that are informative and **efficient**

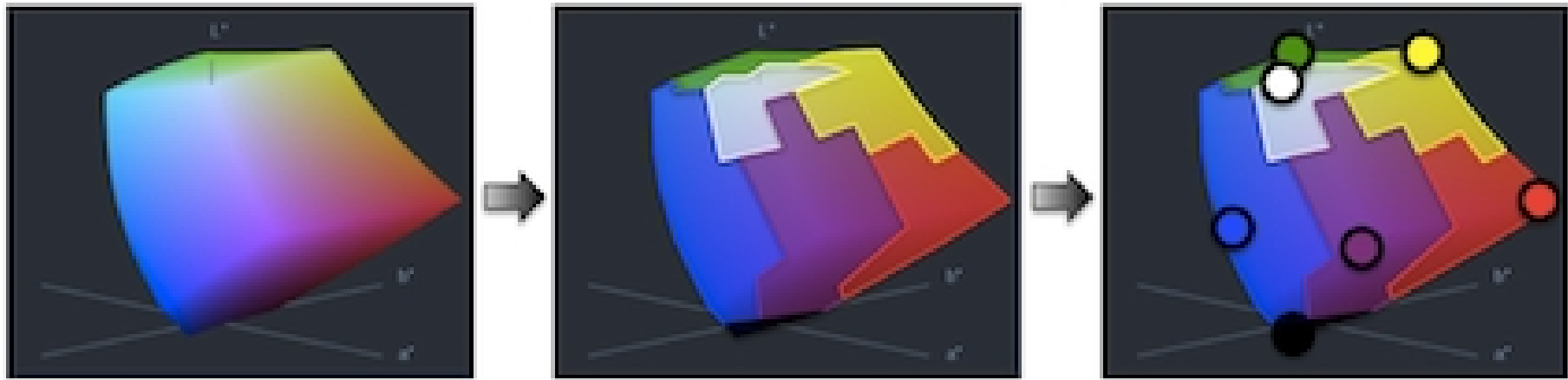


Once those categories have been determined, their best examples can be derived from them as maximally **representative** members of the category (Abbott et al. 2016)



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$$R(c, t) = \log \frac{p(c \mid t)}{p(c \mid \neg t)}$$

chip
term

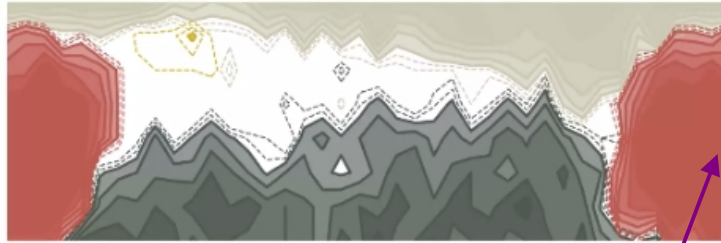
- Estimate $p(c|t)$ from WCS **naming** data

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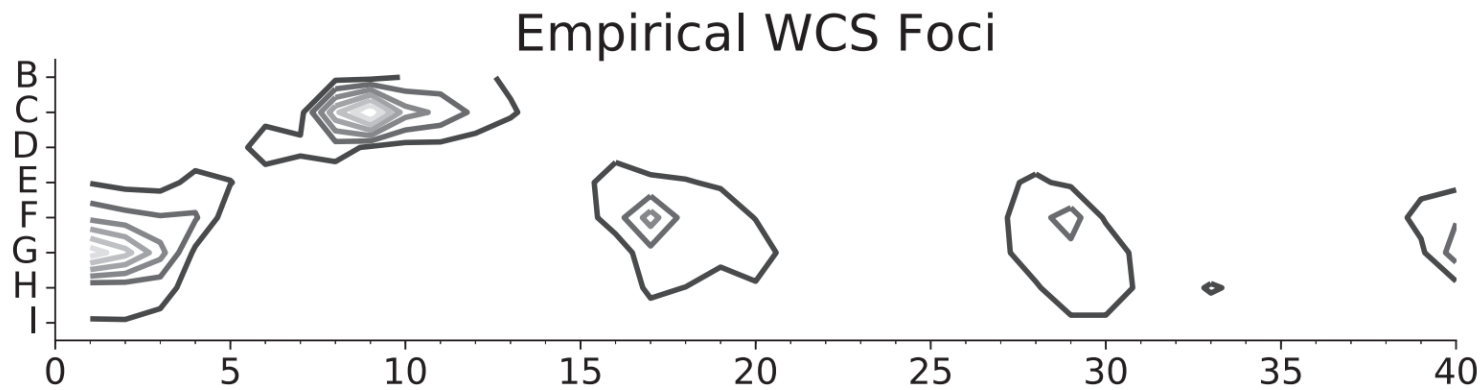
E.g. what is the probability of **this** specific chip c , given that we are considering the reddish term t ?

- Estimate $p(c|t)$ from WCS **naming** data
- Use $R(c,t)$ to predict WCS **focus** data.

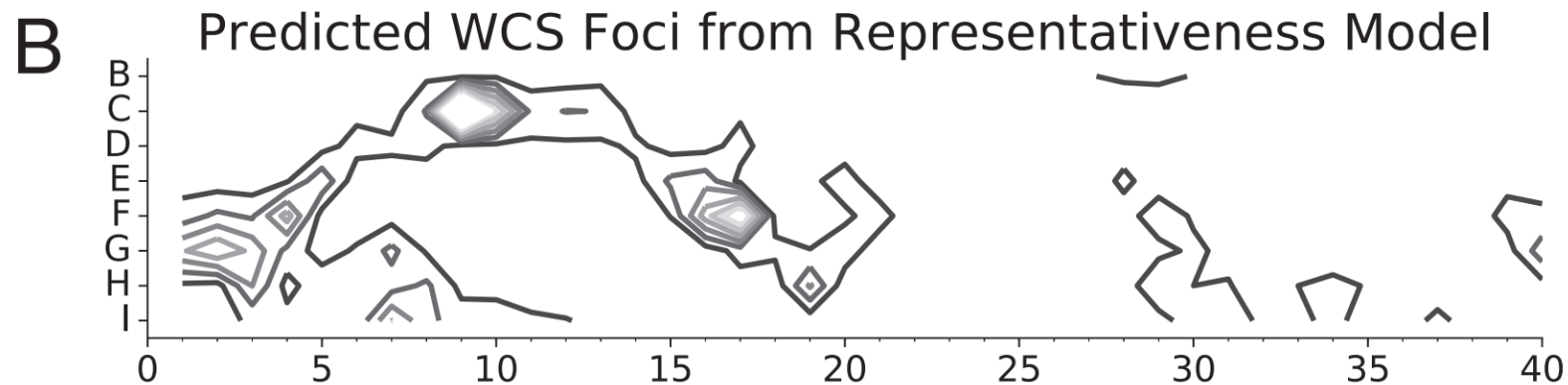
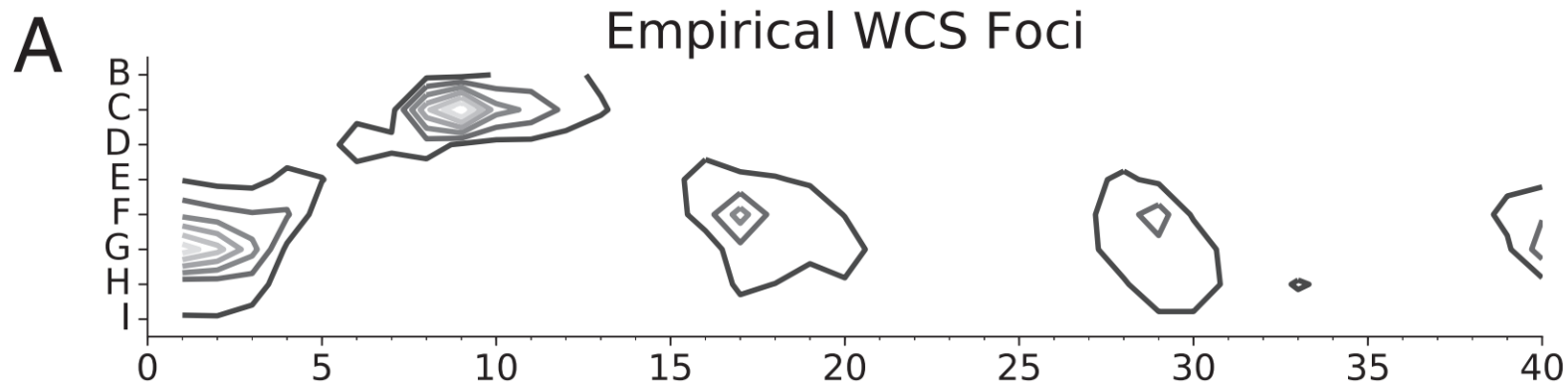


E.g. what is the predicted **best example** of the reddish term t ?

A

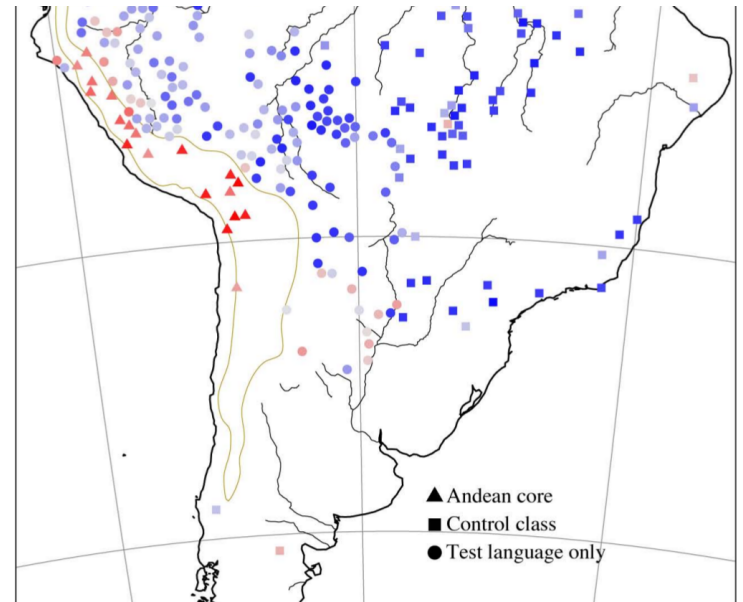


Abbott et al. 2016, *PNAS*

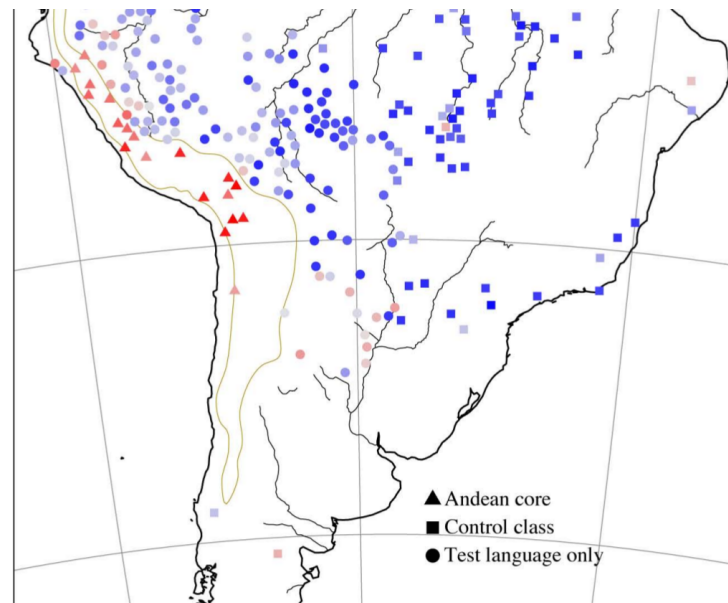


Abbott et al. 2016, *PNAS*

Michael et al. 2014



- Estimate $p(f|A)$, where f is a feature (phoneme) and A is an area (Andean core vs. control) – from SAPHon data, as part of naive Bayes classification.



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- Use $R(f,A)$ to identify which phonemes are **distinctively associated with**, vs. **distinctively absent from**, the Andean core.

TABLE 1

Distinctive features of the Andean core languages. Left: distinctive phonemes (positive feature deltas). Right: distinctive absences (negative feature deltas).

$p^h p'$ $t^h t'$ $k^h k'$ $q q^h q'$ $tf tf^h tf'$ $tʂ$ x χ s $\int$ $l r$ $\mathfrak{p}$ Λ	d k^w $ʔ$ $dʒ$ βf h η^w
$i:$ $u u:$ $a:$	$\tilde{i} \tilde{i}:$ $\dot{i} \dot{i}:$ $\tilde{i} \tilde{i}:$ $\tilde{u} \tilde{u}:$ tone $e \tilde{e} \tilde{e}:$ $\varepsilon \tilde{\varepsilon}$ $\mathfrak{a} \mathfrak{a}:$ $\tilde{\mathfrak{a}} \tilde{\mathfrak{a}}:$ $o \tilde{o} \tilde{o}:$ $\mathfrak{c} \tilde{\mathfrak{c}} \mathfrak{x}$ $\tilde{a}$

Questions? Comments?

$$P(y \mid x_1, \dots, x_n) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$

hey	... 1829
i'm	... 1710
no	... 1191
where	... 1012
you	... 985
speak	... 873
learn	... 747
one	... 739

good letters

viagra	... 1552
casino	... 1492
100%	... 1320
credit	... 1184
sale	... 985
press	... 873
free	... 747
enlarge	... 739

spam letters

«KITTY»

672 times

13 times

THE SIMPLEST SPAM-FILTER
(used until 2010)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

BAYES' THEOREM

✓
NOT SPAM

NAIVE BAYES



FIGURE 2 Languages of South America (two-way Andean core NBC scores)